

Notes: Fisman and Love (JF, 2003)
Trade Credit, Financial Intermediary Development, and Industry Growth

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1 One-sentence summary

Fisman and Love (2003) show that industries that naturally rely more on trade credit grow relatively faster in countries with weaker financial intermediary development, which implies that supplier-provided credit can partly substitute for formal bank finance, especially for the expansion of existing firms rather than the creation of new firms.

2 Big picture

The paper fits into the finance-and-growth literature. Rajan and Zingales (1998) show that industries dependent on external finance grow faster in countries with more developed financial markets. Fisman and Love ask what happens when formal finance is weak. Do firms simply fail to grow, or can they use alternative financing channels?

Their answer is that trade credit - implicit financing from suppliers through accounts payable - can serve as a second-best substitute for bank finance. The basic idea is simple: if banks do not lend easily, suppliers may still extend credit because they have informational, enforcement, liquidation, or relationship advantages over banks.

The key empirical prediction is an interaction: industries with high natural trade-credit use should be less hurt by weak formal financial development. Put differently, trade credit should matter most where financial intermediaries are least developed.

3 Research question

The paper asks four related questions.

1. **Does trade credit substitute for formal finance?** The core test is whether trade-credit-intensive industries grow faster in countries with weak financial intermediaries.
2. **Is this channel distinct from external financial dependence?** The authors test whether the trade credit interaction survives after controlling for the Rajan-Zingales external-finance channel.
3. **Which part of growth does trade credit support?** They decompose industry growth into growth in average establishment size and growth in the number of establishments.
4. **Are the results robust to alternative measurement?** The paper uses stock and flow measures of trade credit, alternative financial development proxies, alternative time periods, small-firm and young-firm samples, instrumental variables, and accounting standards.

4 Incremental contribution

4.1 Extending the finance-and-growth mechanism

The paper builds directly on Rajan and Zingales (1998), but instead of asking only whether financial development helps externally dependent industries, it asks whether an informal financing

mechanism can soften the cost of weak financial development.

This is an important increment because it shifts the question from “does finance matter?” to “what substitutes for finance when finance is underdeveloped?” The paper therefore adds a microeconomic mechanism to the cross-country growth literature.

4.2 Using industry trade-credit intensity as an exogenous benchmark

The empirical challenge is that actual trade credit use in a country is endogenous to local institutions. To avoid this, the authors use U.S. Compustat data to measure each industry’s natural reliance on trade credit. The United States is treated as a benchmark economy where firms’ trade credit choices are closer to technologically or commercially desired levels.

The identifying assumption is similar to Rajan and Zingales: industry characteristics measured in the United States proxy for technological or institutional needs that apply across countries. The paper then asks whether these industry characteristics interact with country-level financial development.

4.3 Separating trade credit from external financial dependence

The paper constructs both a trade-credit intensity measure and a stock measure of external-finance dependence. This allows it to show that the trade credit channel is not just another way of capturing the same external-finance dependence studied by Rajan and Zingales.

The central result survives when both interaction terms are included. Thus, trade credit is not merely a proxy for industries that need outside finance; it has separate explanatory power.

4.4 Growth margin contribution

A particularly useful contribution is the decomposition of industry growth into growth in average establishment size and growth in the number of establishments. Trade credit helps explain growth in the size of existing firms, but not growth in the number of firms. This is consistent with reputation-based trade credit theories: established firms can borrow from suppliers, while new firms have not yet built relationships and credit histories.

5 Theories of trade credit provision

5.1 Comparative advantage in liquidation

Suppliers may lend when banks do not because they can repossess, reuse, or resell the goods they supplied more effectively than financial intermediaries. This argument is strongest for industries that use standardized, easy-to-resell inputs or hold large inventories of raw materials.

Under this view, trade credit intensity differs across industries because products differ in resale value, depreciation, and firm specificity.

5.2 Supplier price discrimination

Trade credit can also be a pricing tool. If suppliers have market power and credit customers have different demand elasticities than cash customers, suppliers can use credit terms to discriminate across buyers. This mechanism can also have industry-level components because input market concentration and product market structure vary by industry.

5.3 Product quality guarantees

Trade credit can act as a warranty. By allowing the buyer to pay later, the supplier gives the buyer time to inspect the quality of inputs. This channel is especially relevant when product quality is difficult to verify at delivery or when products are technologically complex.

5.4 Customized inputs and relationship surplus

When supplier-customer relationships involve customized products, learning-by-doing, or other sunk costs, the relationship generates surplus that rises with its duration. A supplier may extend credit because repeated interaction gives it enforcement power: defaulting today can destroy a valuable future relationship.

5.5 Why trade credit may work when bank finance is weak

Weak legal enforcement and poor information should hurt both banks and suppliers. However, suppliers may still be better positioned than banks for three reasons. First, they obtain information through normal business dealings. Second, they may have better liquidation or resale options. Third, they can enforce repayment through continued supply relationships and relationship-specific surplus.

This is why the paper treats trade credit as a weak substitute for formal finance, not as a perfect substitute.

6 Data and measurement

6.1 Country-industry growth data

The dependent variable is real growth in value added for each industry-country cell from 1980 to 1990. The data come from Rajan and Zingales (1998), ultimately based on United Nations industrial statistics. The sample covers 37 industries and 44 countries, though regressions have slightly fewer observations depending on variable availability.

The regressions include country and industry fixed effects. This means the results are identified from whether the same industry grows differently across countries with different financial development, and whether the same country shows different growth across industries with different trade-credit intensity.

6.2 Trade credit intensity

The main trade credit measure is

$$APAY/TA_i = \text{industry median} \left(\frac{\text{accounts payable}}{\text{total assets}} \right),$$

computed using U.S. Compustat firms. It measures the share of assets financed by supplier credit.

The main flow analogue is

$$\Delta APAY / \Delta TA_i,$$

which measures the proportion of new assets over the decade that are financed by increases in payables.

The authors also use accounts payable scaled by total liabilities, and they recompute industry measures using different decades, small firms, and young firms.

6.3 External finance dependence

To compare with Rajan and Zingales, the paper constructs a stock measure of external finance dependence:

$$EXTFIN_i = \text{industry median} \left(\frac{\text{total assets} - \text{retained earnings}}{\text{total assets}} \right).$$

This measures the fraction of assets not financed by accumulated retained earnings. It is closely related to the Rajan-Zingales flow measure of external finance, but is more comparable to the stock measure of trade credit.

6.4 Financial development

The main country-level measure is private credit to GDP:

$$PRIV_c = \frac{\text{private domestic credit by monetary authorities and deposit institutions}}{GDP}.$$

The authors focus on private credit because trade credit is meant to substitute primarily for bank and intermediary credit, not for stock market financing or politically allocated public credit.

They also test total domestic credit, stock market capitalization, the Rajan-Zingales aggregate financial development measure, GDP per capita, legal-origin instruments, and accounting standards.

7 Empirical specification and identification

7.1 Baseline regression

The central regression is

$$GROWTH_{ci} = \alpha_i + \xi_c + \beta (PRIV_c \times APAY/TA_i) + \varepsilon_{ci},$$

where c indexes countries and i indexes industries.

The paper predicts

$$\beta < 0.$$

A negative coefficient means that trade-credit-intensive industries grow relatively faster when $PRIV$ is low. Equivalently, the advantage of trade credit declines as formal financial intermediaries become more developed.

7.2 Interpretation of the interaction

The coefficient should not be interpreted as saying that trade credit is bad in countries with strong financial systems. Instead, the interaction asks whether trade credit is especially useful when formal finance is weak. If banks are already well developed, industries do not need to rely as much on supplier credit as a substitute.

7.3 Relation to Rajan and Zingales

The Rajan-Zingales prediction is that financially dependent industries grow faster in financially developed countries:

$$GROWTH_{ci} = \alpha_i + \xi_c + \theta (FD_c \times FINDEP_i) + \varepsilon_{ci}, \quad \theta > 0.$$

Fisman and Love include both $APAY/TA_i \times PRIV_c$ and $EXTFIN_i \times PRIV_c$. The trade credit term is negative, while the external finance term is positive. This is exactly the pattern expected if formal finance directly helps financially dependent industries but trade credit substitutes for formal finance when it is weak.

8 Main findings

8.1 Trade-credit-intensive industries grow faster when formal finance is weaker

The main coefficient on $APAY/TA \times PRIV$ is negative and statistically significant. The economic magnitude is meaningful: moving from a financially developed country such as Korea to a weaker financial system such as Egypt widens the growth gap between high- and low-trade-credit industries by about 1.2 percentage points.

The effect is comparable in size to the Rajan-Zingales external-finance interaction. Thus, trade credit is not only statistically detectable; it is economically meaningful.

8.2 The result is independent of external finance dependence

When the authors include $EXTFIN \times PRIV$ and $APAY/TA \times PRIV$ in the same regression, both are significant with the expected signs. Industries that need outside finance benefit from financial development, while industries that can use trade credit do relatively better when financial development is weak.

This is the paper's strongest evidence that trade credit is a distinct channel.

8.3 Private credit is the relevant formal-finance substitute

The results are strongest when financial development is measured by private credit or domestic credit. The stock-market-capitalization interaction is weaker. When private credit and stock market capitalization enter together, the private credit channel dominates.

This is consistent with the interpretation that supplier credit substitutes primarily for bank or intermediary lending rather than public equity financing.

8.4 Trade credit supports existing-firm expansion, not entry

When growth is decomposed, the trade credit interaction predicts growth in average establishment size, but not growth in the number of establishments. This suggests that trade credit mainly helps existing firms expand.

This pattern makes sense because trade credit depends on relationships, trust, and reputation. New firms have less ability to obtain supplier credit because they have not established histories with suppliers.

9 Robustness checks

9.1 Alternative financial development measures

The paper uses multiple measures: private credit, total domestic credit, stock market capitalization, and the Rajan-Zingales financial development measure. The negative trade credit interaction remains robust, especially for credit-based financial development measures.

9.2 Flow measures of trade credit and external finance

The paper reproduces the main results using $\Delta APAY/\Delta TA$ and the original Rajan-Zingales flow measure of external finance. The coefficient on the trade credit interaction remains negative and significant.

This matters because the main specification uses stock variables. The flow specification shows that the result is not an artifact of the stock measure.

9.3 Alternative denominators and time periods

The authors use accounts payable over total liabilities instead of accounts payable over total assets, and they calculate the industry measures using data from the 1970s and the 1990s. The results remain similar. This reinforces the claim that trade credit use is a stable industry characteristic rather than a period-specific artifact.

9.4 Endogeneity of financial development

A reverse-causality concern is that countries with more trade-credit-suitable industries may have less need to develop formal credit markets. The paper addresses this by instrumenting financial development with legal origin, following Rajan and Zingales. The result remains negative and significant, although weaker.

9.5 Accounting standards

The paper also uses accounting standards as an alternative proxy for the information environment available to financial intermediaries. The interaction of trade credit intensity with accounting standards is negative and weakly significant. This is consistent with the view that when public accounting information is poor, suppliers may have a relative informational advantage.

10 Interpretation

The paper’s interpretation is not that trade credit is superior to bank credit. Instead, supplier credit is a second-best workaround in economies where banks and formal intermediaries are underdeveloped.

The finding that trade credit does not predict entry is important. It suggests that trade credit is less democratic than formal finance. It helps firms that already have supplier relationships, but it may not help new entrepreneurs enter markets. Thus, trade credit can partly offset weak financial development, but it cannot fully replace well-functioning financial intermediaries.

11 Limitations

The key identification assumption is that U.S. industry trade-credit intensity reflects a technological or commercial characteristic that applies across countries. This is plausible but strong. Industries may use different production chains, contract terms, or supplier relationships in different countries.

The analysis is also cross-sectional at the country-industry level, so it cannot directly observe individual firm borrowing constraints or contract terms. The results are best interpreted as reduced-form evidence that trade credit can substitute for formal finance in the aggregate pattern of industry growth.

Finally, the paper does not identify which specific theory of trade credit is most important. The empirical measure captures overall trade-credit suitability, which may reflect liquidation value, quality verification, price discrimination, relationship-specific surplus, or a combination of all of these.

12 Tables and figures

12.1 Table I: List of Variable Abbreviations, Definitions, and Sources

12.2 Table II: Trade Credit Use by Industry

Abbreviation	Description
<i>1. Industry-level variables, original source Compustat:</i>	
$\Delta EXTFIN$	Dependence on external financing, industry-level median of the ratio of capital expenditures minus cash flow over capital expenditures (the numerator and denominator are summed over all years for each firm before dividing). This variable measures the portion of capital expenditures not financed by internally generated cash. Cash flow is defined as the sum of funds from operations, decreases in inventories, decreases in receivables, and increases in payables. From Rajan and Zingales (1998).
$EXTFIN$	Stock measure of external finance, equal to the ratio of total assets minus retained earnings over total assets, industry medians of ratios over all firm-years in the relevant time period. This variable measures the accumulated stock of internally generated funds.
$APAY/TA$	Stock measure of dependence on trade credit, equal to accounts payable over total assets, industry medians of ratios over all firm-years in the relevant time period.
$\Delta APAY/ATA$	Flow measure of dependence on trade credit, equal to the change in accounts payable over change in total assets over the decade, industry medians of firm-level measures (the numerator and denominator are summed over all years for each firm before dividing, similarly to $\Delta EXTFIN$).
$APAY/TL$	Accounts payable scaled by total liabilities, industry medians of firm-level measures.
<i>2. Country-industry level variables from Rajan and Zingales (1998); originally from United Nations Statistics:</i>	
Industry growth	Main measure is growth in value added; it is annual compounded growth rate in real value added estimated for the period 1980 to 1990 for each ISIC industry in each country. Growth in number equals to the difference in the log of ending period firm number less the log of firm number in the beginning of period. Growth in average size equals the difference in logs of the average size, which is defined as total value added in the industry divided over the number of firms in the industry.
Fraction	Fraction of the industry's value added in total manufacturing value added for 1980.
<i>3. Country-level variables from Rajan and Zingales (1998):</i>	
Domestic credit	Ratio of domestic credit held by monetary authorities and depositary institutions (excluding interbank deposits) scaled by GDP for 1980. Original source is International Financial Statistics (IFS).
Private credit	Ratio of private domestic credit held by monetary authorities and depositary institutions (excluding interbank deposits) scaled by GDP for 1980. Original source is IFS.
Market cap	Ratio of stock market capitalization to GDP in 1980. Original source is IFS.
Log GDP PC	Log of GDP per capita in U.S. dollars in 1980. Original source is IFS.
Legal origin	Dummies for English, French, German, or Scandinavian origin of the legal system (La Porta et al. (1996)).
Accounting standards	Amount of disclosure of companies' annual reports in each country (La Porta et al (1996)).

Industry medians of firm-level variables. See Table I for variable definitions and sources.				
ISIC code	Description	APAY/TA 1980s	$\Delta APAY/ATA$	EXTFIN 1980s
311	Food Products	11.2%	5.7%	72%
313	Beverages	9.1%	5.3%	61%
314	Tobacco	6.6%	4.0%	63%
321	Textile	10.1%	7.5%	68%
3211	Spinning	14.9%	17.5%	69%
322	Apparel	11.1%	7.9%	66%
323	Leather	5.5%	2.0%	64%
324	Footwear	9.3%	4.3%	59%
331	Wood Products	8.8%	8.4%	75%
332	Furniture	9.2%	4.8%	64%
341	Paper and Products	8.1%	5.6%	67%
3411	Pulp, Paper	6.5%	6.1%	65%
342	Printing and Publishing	7.5%	5.3%	64%
3511	Basic Chemicals excl. Fertil.	8.3%	5.6%	68%
3513	Synthetic Resins	9.3%	7.3%	74%
352	Other Chemicals	9.7%	6.7%	63%
3522	Drugs	5.5%	3.1%	94%
353	Petroleum Refineries	11.8%	21.7%	69%
354	Petroleum and Coal Products	9.6%	7.0%	65%
355	Rubber Products	8.8%	13.4%	66%
356	Plastic Products	9.9%	9.7%	80%
361	Pottery	6.7%	3.2%	66%
362	Glass	8.9%	3.5%	65%
369	Non-metal Products	6.4%	6.8%	71%
371	Iron and Steel	9.4%	9.1%	72%
372	Non-Ferrous Metal	7.8%	8.1%	73%
381	Metal Products	8.8%	8.0%	71%
382	Machinery	8.6%	8.5%	73%
3825	Office, Computing	8.3%	5.7%	89%
383	Electric Machinery	8.2%	7.7%	80%
3832	Radio	7.6%	6.6%	83%
384	Transportation Equipment	10.5%	5.9%	71%
3841	Ship	10.1%	7.9%	83%
3843	Motor Vehicle	11.2%	9.9%	75%
385	Professional Goods	7.2%	4.9%	86%
390	Other ind.	8.7%	7.8%	77%
	Mean	8.9%	7.3%	71%
	Median	8.8%	6.8%	70%
	Standard deviation	1.9%	3.8%	8%

Table I: List of Variable Abbreviations, Definitions, and Sources

Table II: Trade Credit Use by Industry

Read for Table I: Table I defines the main variables. The key industry-level variable is $APAY/TA$, accounts payable over total assets, which captures an industry's dependence on trade credit. The main country-level financial development variable is $PRIV$, private domestic credit over GDP. The paper's core identification comes from interacting an industry variable with a country variable.

Read for Table II: Table II shows that trade credit use varies substantially across industries, from roughly 5 percent to 15 percent of assets. Petroleum refineries, motor vehicles, and spinning are among high trade-credit industries, while drugs, leather, and pottery are among low trade-credit industries. This cross-industry variation is the empirical basis for the paper.

12.3 Table III: Correlations of Industry-Level Measures

12.4 Table IV: Accounts Payable and Industry Growth

Table III
Correlations of Industry-Level Measures

Correlations of industry-level medians. See Table I for variable definition and sources. All correlations are based on 36 observations, except $\Delta EXTFIN$ measured over 1970s, which has only 35 observations. P -values are in parenthesis (0 stands for lower than 0.01). * denotes significance at one percent.

	APAY/TA 1980's	APAY/TA 1970's	$\Delta APAY/TA$ 1980's	$\Delta APAY/TA$ 1970's	EXTFIN 1980's	EXTFIN 1970's	$\Delta EXTFIN$ 1980's
APAY/TA 1970's	0.7529* (0.00)						
$\Delta APAY/TA$ 1980's	0.6725* (0.00)	0.5888* (0.00)					
$\Delta APAY/TA$ 1970's	0.4648* (0.00)	0.5175* (0.00)	0.4339* (0.01)				
EXTFIN 1980's	-0.1328 (0.44)	0.1475 (0.39)	0.0152 (0.93)	-0.0706 (0.68)			
EXTFIN 1970's	0.303 (0.07)	0.4553* (0.01)	0.2796 (0.10)	0.2092 (0.22)	0.6185* (0.00)		
$\Delta EXTFIN$ 1980's	-0.1604 (0.35)	0.1023 (0.55)	-0.11 (0.52)	-0.0282 (0.87)	0.8213* (0.00)	0.4362* (0.01)	
$\Delta EXTFIN$ 1970's	-0.0121 (0.95)	0.1181 (0.50)	0.1107 (0.53)	0.0555 (0.75)	0.6748* (0.00)	0.7415* (0.00)	0.6292 (0.00)

Table III: Correlations of Industry-Level Measures

Read for Table III: Table III supports the idea that trade credit intensity is persistent across time. The correlation of $APAY/TA$ between the 1970s and 1980s is high, and the flow measure is also positively correlated with the stock measure. This persistence makes it more plausible that trade credit intensity is an industry characteristic rather than pure noise.

Read for Table IV: Table IV is the main result. The interaction $APAY/TA \times PRIV$ is negative and significant, meaning trade-credit-intensive industries grow relatively faster where private credit is less developed. Including $EXTFIN \times PRIV$ leaves both channels significant, showing that the trade credit mechanism is distinct from the external-finance mechanism.

Table IV
Accounts Payable and Industry Growth

The dependent variable is real growth in value added. Fraction is fraction of industry value added in total manufacturing in 1980; $APAY/TA$ is industry's dependence on trade credit measured by accounts payable over total assets; $EXTFIN$ is stock measure of external finance, equal to total assets minus retained earnings over total assets. Domestic credit is the ratio of total credit to GDP in 1980; $PRIV$ is financial intermediary development in 1980 (the ratio of total credit held by private (nongovernmental) organizations to GDP). All variables are from RZ except for $APAY/TA$ and $EXTFIN$, which are from COMPUSTAT 1980 to 1989. All models except Model 5 include country and industry dummies. Standard errors appear in parentheses, and are adjusted for heteroskedasticity. Significance levels ***, **, and * correspond to 1 percent, 5 percent and 10 percent, respectively.

	1	2	3	4	5
Fraction	-0.85 (0.246)***	-0.88 (0.247)***		-0.95 (0.248)***	-0.622 (0.187)***
$EXTFIN \times PRIV$	0.295 (0.150)**			0.307 (0.151)**	
$APAY/TA \times PRIV$		-2.01 (0.597)***	-1.64 (0.604)***	-2.06 (0.600)***	-2.17 (0.720)***
$APAY/TA$					0.84 (0.370)**
log GDP PC					-0.013 (0.005)***
$PRIV$					0.277 (0.067)
N	1,217	1,217	1,217	1,217	1,171
R^2	0.28	0.29	0.27	0.30	0.065

Table IV: Accounts Payable and Industry Growth

12.5 Table V: Composition of Capital Markets and GDP

12.6 Table VI: Growth in Number and Size

Table V Composition of Capital Markets and GDP						
The dependent variable is real growth in value added. Fraction is fraction of industry value added in total manufacturing in 1980; APAY/TA is industry's dependence on trade credit measured by accounts payable over total assets, EXTFIN is stock measure of external finance, equal to total assets minus retained earnings over total assets. Domestic credit is the ratio of total credit to GDP in 1980; PRIV is financial intermediary development in 1980 (the ratio of total credit held by private (nongovernmental) organizations to GDP); Market cap is the ratio of stock market capitalization to GDP in 1980; FD is the financial development measure used by RZ (the sum of domestic credit and market capitalization). All variables are from RZ except for APAY/TA and EXTFIN, which are from COMPUSTAT 1980-1989. All models include country and industry dummies. Standard errors appear in parentheses, and are adjusted for heteroskedasticity. Significance levels ***, **, and * correspond to 1 percent, 5 percent and 10 percent, respectively.						
	1	2	3	4	5	6
Fraction	-0.91 (0.25)***	-0.85 (0.25)***	-0.87 (0.25)***	-0.93 (0.25)***	-0.91 (0.25)***	-0.92 (0.25)**
Interaction of APAY/TA with:						
FD	-1.12 (0.38)***					
Domestic credit		-1.42 (0.51)***				
PRIV				-1.83 (0.57)***	0.68 (0.71)	-1.42 (0.56)**
Market Cap			-0.76 (0.35)***	-0.49 (0.29)*	0.68 (0.60)***	
Log GDP PC						-0.19 (0.14)
Interaction of EXTFIN with:						
FD	0.18 (0.08)***					
Domestic credit		0.15 (0.13)			-0.15 (0.13)	
PRIV				0.32 (0.16)**	0.22 (0.18)	
Market Cap			0.15 (0.08)**	0.12 (0.08)		
Log GDP PC						0.025 (0.02)
N	1,217	1,217	1,217	1,217	1,217	1,217
R ²	0.29	0.29	0.28	0.29	0.29	0.29

Table V: Composition of Capital Markets and GDP

Table VI Growth in Number and Size							
The dependent variable is growth in the average size of establishments in Models 1 and 2 and growth in the number of establishments in Models 3-8. Fraction is fraction of industry value added in total manufacturing in 1980; APAY/TA is industry's dependence on trade credit measured by accounts payable over total assets; EXTFIN is stock measure of external finance, equal to total assets minus retained earnings over total assets. PRIV is financial intermediary development in 1980 (the ratio of total credit held by private (nongovernmental) organizations to GDP). Models 5 and 6 use industry variables calculated for small firms only (less than 25th percentile for each industry in real sales), and Models 7 and 8 use industry variables calculated for young firms only (less than 10 years after IPO at the start of the period). All models include country and industry dummies. Standard errors appear in parentheses, and are adjusted for heteroskedasticity. Significance levels *** **, and * correspond to 1 percent, 5 percent and 10 percent, respectively.							
Dependent variable	Growth in average size of establishments		Growth in number of establishments				
	1	2	All firms	Small Firms	Young Firms		
Fraction	-0.77 (0.33)**	-0.78 (0.33)**	-0.19 (0.14)	-0.2 (0.14)**	-0.19 (0.14)	-0.19 (0.14)	-0 (0.14)
APAY/TA*PRIV	-1.97 (0.58)***	-1.89 (0.57)***	-0.47 (0.38)	-0.42 (0.36)	-0.32 (0.33)	-0.39 (0.34)	-0.4 (0.36)
EXTFIN*PRIV			0.15 (0.13)	0.13 (0.097)	0.064 (0.047)	0.064 (0.047)	0.17 (0.06)
N	1,047	1,047	1,073	1,073	1,073	1,073	1,073
R ²	0.43	0.42	0.45	0.45	0.45	0.45	0.44

Table VI: Growth in Number and Size

Read for Table V: Table V shows that the substitutability result is strongest for credit-based measures of financial development. The interaction with stock market capitalization is weaker, and private credit remains the more relevant substitute for trade credit. This is consistent with the paper's view that supplier credit substitutes mainly for bank-type finance.

Read for Table VI: Table VI decomposes growth into growth in average establishment size and growth in the number of establishments. Trade credit predicts growth in average size but not the number of establishments. This supports reputation-based theories: supplier credit helps existing firms expand, but it does not substantially help new firms enter.

12.7 Table VII: Flow Measures of Trade Credit and External Finance

12.8 Table VIII: Alternative Measures

Table VII Flow Measures of Trade Credit and External Finance								
Fraction is fraction of industry value added in total manufacturing in 1980 and $\Delta APAY/\Delta TA$ is the flow measure for industry's dependence on trade credit measured by change in accounts payable over change in total assets. $\Delta EXTFIN$ is the flow measure of external finance equal financial dependence measure from RZ. PRIV is financial intermediary development in 1980 (the ratio of total credit held by private (nongovernmental) organizations to GDP); FD is financial development (the sum of domestic credit and market capitalization). All variables are from RZ except for $\Delta APAY/\Delta TA$, which is from COMPUSTAT 1980 to 1989. All models include country and industry dummies. Model 7 uses industry variable calculated for small firms only (less than 25th percentile for each industry in real sales), and model 8 uses industry variables calculated for young firms only (less than 10 years after IPO at the start of the period). Standard errors appear in parentheses, and are adjusted for heteroskedasticity. Significance levels *** **, and * correspond to 1 percent, 5 percent and 10 percent, respectively.								
Dependent variable:	1	2	3	4	5	6	7	8
	Total growth				Growth in size	Growth in number of establishments		
Fraction	-0.84 (0.240)***	-0.91 (0.24)***	-0.92 (0.24)***	-0.96 (0.24)***	-0.77 (0.33)***	-0.2 (0.15)	-0.2 (0.14)	-0.14 (0.16)
Interaction of $\Delta APAY/\Delta TA$ with:								
PRIV	-0.85 (0.29)***	-0.77 (0.29)***			-0.76 (0.34)**	0.03 (0.18)	-0.025 (0.04)	-0.07 (0.20)
FD			-0.56 (0.17)***	-0.475 (0.18)***				
Log GDP PC				-0.046 (0.04)				
Interaction of $\Delta EXTFIN$ with:								
PRIV		0.11 (0.036)***			0.058 (0.022)**	0.038 (0.018)**	0.039 (0.019)**	0.025 (0.013)**
FD			0.07 (0.022)***	0.051 (0.025)**				
Log GDP PC				0.012 (0.008)				
N	1,217	1,217	1,217	1,217	1,047	1,979	1,073	1,073
R ²	0.28	0.29	0.28	0.29	0.42	0.45	0.45	0.44

Table VII: Flow Measures of Trade Credit and External Finance

Table VIII Alternative Measures		1	2	3	4	5	6	7	Accounti									
Fraction is fraction of industry value added in total manufacturing in 1980, and APAY/TA is industry's dependence on trade credit measured by accounts payable over total assets. EXTFIN is stock measure of external finance, equal to total assets minus retained earnings over total assets. PRIV is financial intermediary development in 1980 (the ratio of total credit held by private (nongovernmental) organizations to GDP). Models 1 and 2 use APAY/TA measured over 1980s; Models 3 and 4 use APAY/TA and EXTFIN measured over 1970s; Models 5 and 6 use APAY/TA measured over 1990s; Model 7 is estimated by instrumental variables with legal origin used as instruments for PRIV - financial intermediary development. Model 8 is the interaction of APAY/TA with accounting standards. All models include country and industry dummies. Standard errors appear in parentheses, and are adjusted for heteroskedasticity. Significance levels ***, **, and * correspond to 1 percent, 5 percent and 10 percent, respectively.																		

Table VIII: Alternative Measures

Read for Table VII: Table VII repeats the main tests using flow measures. The trade credit interaction remains negative and significant for total growth and for growth in average size, but not for the growth in the number of establishments. This confirms that the main results are not an artifact of using stock measures.

Read for Table VIII: Table VIII shows robustness to alternative trade credit denominators, different decades for measuring industry characteristics, instrumental variables, and accounting standards. The negative trade credit interaction remains, which strengthens the interpretation that trade credit substitutes for weak formal finance.

13 Key takeaways

13.1 Core conclusion

Trade credit matters for growth because it can substitute for formal financial intermediation. Industries that naturally use more supplier credit are relatively less constrained in countries where banks and other formal intermediaries are weak.

13.2 Economic intuition

Suppliers can sometimes lend when banks cannot because they know their customers, can threaten future supply relationships, and may be better able to repossess or resell the goods they supplied. These advantages are industry-specific, which creates the variation used in the empirical design.

13.3 Important nuance

Trade credit is not a perfect replacement for financial development. It mostly helps existing firms expand, not new firms enter. Therefore, weak financial systems may still reduce entrepreneurship and competition even when supplier credit mitigates constraints for incumbent firms.

13.4 Why the paper is influential

The paper provides a clean example of how firms adapt to weak institutions. It also shows that informal financing arrangements can have real aggregate consequences, but that their benefits are uneven across industries and firm life-cycle stages.

13.5 Bottom line

Formal financial development promotes growth, but firms are not passive when finance is weak. Trade credit is one important substitute, and its industry variation helps explain why some sectors grow despite underdeveloped financial intermediaries.